

Math 150B – Calculus II

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Office Hours: T 3:00PM – 4:00 PM, TH 12:00PM–1:00PM also available by appointment.

Prerequisites Math 150A (with a grade of “C” or better.)

Textbook:

- “Calculus” volume 2 by OpenStax

Available at <https://openstax.org/details/books/calculus-volume-2>.

Course Description: This course is the second semester in a three-semester sequence in calculus. Topics covered include techniques of integration, improper integrals, applications of integration, an introduction to differential equations, parametric equations, and sequences and series.

MyOpenMath Assignments: There will be weekly homework assigned through MyOpenMath online which can be accessed through the class Canvas.

Quizzes/Classwork: There will be weekly quizzes or classwork based on recent lectures.

Important dates:

- September 9 – Last day for students to ADD with a permit. All permits expire at midnight on September 5.
- September 9 – Last day for students to DROP without grade of “W”. Students drop using Titan Online
- September 19 – Last day the Math Department will be flexible on the approval of non-medical withdrawal requests. Beginning Monday, September 22, students must have a serious and compelling reason for non-medical withdrawal requests and must provide supporting documentation for their reason.
- November 14 – Last day to withdraw with a truly serious and compelling reason that is clearly beyond the student’s control. Students must document their reason.
- December 5 – Last day to withdraw for medical reasons. Students must document their reason.

Other Items: <https://fdc.fullerton.edu/teaching/student-info-syllabi.html> For information for students with special needs, emergency preparedness, and more, go to the link in the topic bar.

Midterm Exams: There will be 3 midterms during class time. The tentative dates for the midterm are

- Exam 1 – September 18, 2025
- Exam 2 – October 16, 2025
- Exam 3 – November 20, 2025

Final Examination Date: The schedule for the final exam is available through the student portal. This date is not flexible and the final is CUMULATIVE.

Grades: Letter grades will be determined by the standard 10 point scale. I may use the following qualitative factors to raise or lower your class percentage by 2-3 (basically, to deal with borderline grade cases): attendance, participation in class discussions, hard work, improvement, office hours, etc....

The breakdown of your final grade in the course will be as follows:

Quizzes/Classwork	5%
Homework	20%
Midterms	15% each
Final Exam	30%

Academic Integrity: Students who violate university standards of academic integrity are subject to disciplinary sanctions, including failure in the course and suspension from the university. Since dishonesty in any form harms the individual, other students, and the University, policies on academic integrity are strictly enforced. Examples of academic dishonesty include, but are not limited to;

1. copying from another student's homework, quiz or exam,
2. allowing another student to copy your work,
3. copying homework solutions from the text solutions manual.

You should familiarize yourself with the academic integrity guidelines in the current student handbook.

Disability Support Services (DSS): Students have the right to accommodations for documented special needs. Contact the DSS for possible arrangements. www.fullerton.edu/DSS/

Technology requirements: You will need a computer with internet access and webcam.

Calculators: Are not required but may help in some computations. You can use a scientific calculator on an exam, but no graphing or cell phone app. If in doubt, please ask before the exam if your calculator is allowed.

Rescheduling exams/assignments: Please let me know at least a week ahead of time and we can discuss alternative arrangements for exams and assignments.

Student Learning Goals for Math 150B:

1. Understand and successfully use the integration techniques of integration by parts, trigonometric substitution and partial fraction decomposition, and to learn the strategies for identifying which technique may best work in helping to evaluate a given integral.
2. Understand, and be able to use, the Trapezoid, Midpoint and Simpson's Rules to approximate definite integrals (along with error bounds).
3. Correctly identify when a definite integral is improper and be able to determine if it converges or diverges by either direct calculation or by using the comparison test.
4. Have learned how to apply the definite integral and integration techniques to problems involving finding the arc length of a curve, the area of a surface or revolution, moments and centroids, and hydrostatic pressure and force.
5. Understand the concept of a differential equation, and be able to determine if a given function is a solution of a particular differential equation.
6. Understand the concept of a slope field and be able to correctly graph simple examples. Be able to sketch an approximate graph on the solution to an initial value problem in a field, and correctly identify equilibrium solutions.
7. Correctly identify when a DE is separable and be able to set up and solve separable DE's for a variety of applications (such as population growth and mixture problems).
8. Understand the concept of parametric equations, be able to graph parametric curves (including the direction of motion on the curve), and moreover eliminate the parameter (where possible) to find an equation in x and y for the curve.
9. Understand and successfully apply calculus to parametric curves to find derivatives, areas and arc lengths.
10. Understand polar coordinates, be able to graph polar curves and apply calculus to polar curves to find derivatives, areas and arc length.
11. Understand the concepts of sequences and limits of sequences, and be able to calculate the limit of a sequence using a continuous function.
12. Grasp the concept of a series (i.e. understand that its sum is the limit of the sequence of its partial sums), and understand geometric series and conditions for convergence and divergence.
13. Apply series tests (Integral test, p-series test, comparison tests, alternating series test, ratio test) and be able to identify which test may work best in determining if a series converges or diverges.
14. Comprehend the theory of Power series, including radius of convergence and interval of convergence. Be able to develop new power series from known power series.
15. Understand the theory behind Taylor series, and be able to find them by using the definition and known examples.

These goals are achieved through the course work, including homework, classroom activities, quizzes, exams and projects, which require the student to demonstrate understanding of the mathematical concepts presented in the course and to apply these concepts to the solutions of real world applied problems.

Tentative Schedule:

- Week 1 (8/26, 8/28) – 1.3,4,5,6,7, 2.1
- Week 2 (9/2, 9/4) – 2.2,3
- Week 3 (9/9, 9/11) – 2.4,5,6
- Week 4 (9/16, 9/18) – Review, Exam 1
- Week 5 (9/23, 9/25) – 3.1,2,3
- Week 6 (9/30, 10/2) – 3.4,6,7
- Week 7 (10/7, 10/9) – 4.1,2,3,4,5
- Week 8 (10/14, 10/16) – Review, Exam 2
- Week 9 (10/21, 10/23) – 5.1,2,3,4
- Week 10 (10/28, 10/30) – 5.5,6
- Week 11 (11/4, 11/6) – 6.1,2
- Week 12 (11/13) – 6.3,4
- Week 13 (11/18, 11/20) – Review, Exam 3
- Thanksgiving break
- Week 14 (12/2, 12/4) – 7.1,2,3,4
- Week 15 (12/9, 12/11) – Final Review